

plexity when compared to MoE. This shift in approach offers a unique and more straightforward way of working, moving away from conventional MoE challenges. Not only that, DUS also undergoes continued pretraining to quickly recover performance of the scaled model.

B.3 Prompt Engineering

A key research area to harness the emergent abilities of LLMs is prompt engineering. Prompt engineering is the study of how to design inputs (prompts) that enable LLMs to better perform specific tasks. A prime example of this research is Chain-of-Thought (CoT) (Wei et al., 2022b), which proposes CoT prompting that decomposes multi-step problems into a series of intermediate reasoning steps. Moreover, efforts are underway to replace even such prompt engineering with LLMs (Yang et al., 2023).

B.4 Instruction Tuning

To enhance the steerability of LLMs, instruction tuning (Wei et al., 2021) has emerged as a learning technique. This involves fine-tuning LLMs using data formatted as (instruction, input, output) for various tasks (Wang et al., 2022). Instruction tuning allows for targeted adjustments, providing a more controlled and task-oriented improvement to the model's capabilities.

Before instruction tuning, existing methods faced challenges in effectively guiding and controlling the behavior of large language models (Zhang et al., 2023b). The sheer complexity of these models made it difficult to ensure precise and task-oriented responses. The need for a more targeted approach arose from the limitations of existing methods, leading to the development of instruction tuning. This targeted approach enables better control over the model's behavior, making it more suitable for specific tasks and improving its overall performance in alignment with user-defined objectives. Therefore, instruction tuning is computationally efficient and facilitates the rapid adaptation of LLMs to a specific domain without requiring extensive retraining or architectural changes.

B.5 Alignment Tuning

LLM has been observed to generate sentences that may be perceived as linguistically incongruent by human readers since they learned not human intention, but only vast knowledge across various domains in the pretraining step (Ziegler et al., 2019).

To overcome this limitation and align with human intentions, previous research (Ziegler et al., 2019) have proposed Reinforcement Learning with Human Feedback (RLHF). RLHF operates by learning a reward model based on human preferences, employing reinforcement learning to guide the LLM towards prioritizing answers with the highest reward scores. This process enhances the safety, propriety, and overall quality of the generated responses. Despite demonstrating satisfactory performance, RLHF encounters challenges such as managing numerous hyperparameters and necessitating the incorporation of multiple models (policy, value, reward, and reference models).

In response to these challenges, the supervised fine-tuning based approaches have proposed, such as Rank Responses to align Human Feedback (RRHF) (Yuan et al., 2023), Reward rAnked Fine-Tuning (RAFT) (Dong et al., 2023), and Direct Policy Optimization (DPO) (Intel, 2023). They avoid the complexities associated with reinforcement learning while achieving empirical performance comparable to RLHF. Among them, DPO that we used directly guides the LLM to increase the probability of positive responses and decrease the probability of negative responses through a "direct" approach. Interestingly, DPO demonstrates more stable learning results compared to RLHF, despite its simple training approach.

B.6 Data Contamination

Recent researches (Zhou et al., 2023; Sainz et al., 2023; Golchin and Surdeanu, 2023; Deng et al., 2023) emphasize the need to measure whether a specific benchmark was used to train the large language models. There are three types of the data contamination: guideline, raw text and annotation (Sainz et al., 2023). **Guideline contamination** occurs when a model accesses detailed annotation guidelines for a dataset, providing advantages in specific tasks, and its impact should be considered, especially in zero and few-shot evaluations. **Raw text contamination** occurs when a model has access to the original text. Wikipedia is widely used as a pretraining data, but also as a source for creating new datasets. The caution is advised in the development of automatically annotated datasets sourced from the web. **Annotation contamination** occurs when the annotations of the specific benchmark are exposed during model training.